MAATHRUCARE: AN AI-BASED PREGNANCY SUPPORT SYSTEM FOR SRI LANKAN MOTHERS

INDIVIDUAL THESIS

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Individual component: Virtual Nutritionist & GDM Action Engine

Implemented focus: Ingredient-Based Sri Lankan Meal Recommendation and Nutritional Analysis

Group ID: R26-IT-146

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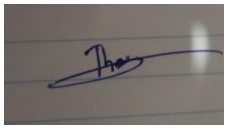
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ABSTRACT

This thesis investigates the nutrition-support contribution to Maathru Care (R26-IT-146), an integrated maternal-health application in Sri Lanka. The work comprises locally contextualised meal assistance and nutritional information, together with a camera-based food-recognition pipeline documented in the supplied React Native/Expo source inspection. Images are compressed on device and sent to a custom YOLOv8 object detector through a FastAPI endpoint; predicted food labels are normalised and matched to nutrient records maintained in Supabase/PostgreSQL. The model-validation screenshot provided by the researcher reports precision 94.32%, recall 94.17%, mAP@50 93.92% and mAP@50–95 92.98%. A class-normalised confusion matrix and a sample prediction montage provide additional evidence of detection behaviour. These measurements concern object detection, not nutritional correctness, clinical safety, or user outcomes. The actual training log, dataset manifest, class frequencies and validation commands were not supplied, preventing independent re-evaluation of the reported scores. The implementation also documents per-portion meal logging and rule-based GDM warnings; the warning thresholds must not be interpreted as validated medical guidance. The research contribution is a local-food-focused nutrition workflow that integrates AI-based recognition, food-composition lookup and meal recording. Remaining limitations, including reference-food mapping and independent nutritional validation, are clearly separated from completed software functions.

In this study, I frame the implemented scope as an ingredient-to-recommendation workflow. The user identifies available ingredients, the module returns meal suggestions suitable for the local dietary context, and associated nutritional information supports comparison and informed selection. A Design Science Research perspective is used to define the problem, describe the artefact and specify how recommendation relevance, ingredient feasibility, nutrition correctness, usability and safety should be evaluated. The available evidence includes a technical code-inspection transcript and YOLO validation figures, but does not establish complete source-level provenance, independent nutrient benchmarks or participant-based usability results. Therefore, this thesis records the confirmed functional scope and a detailed, reproducible validation plan without inventing quantitative performance results or claiming clinical efficacy. The earlier GDM risk engine and other proposal features are treated as planned extensions rather than completed contributions. The central contribution is a culturally focused meal-planning interface linking household ingredients to dietary information; its effectiveness remains to be demonstrated through documented testing.

Keywords: Sri Lankan cuisine, ingredient-based recommendation, nutritional analysis, pregnancy support, meal planning

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CTG	Cardiotocography
DSR	Design Science Research
GDM	Gestational Diabetes Mellitus
GI	Glycaemic Index
GL	Glycaemic Load
LLM	Large Language Model
RDA	Recommended Dietary Allowance
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
SUS	System Usability Scale
UI	User Interface
UX	User Experience

1 INTRODUCTION

1.1 Overall project context

Maathrucare is a group research project that aims to support pregnant women in Sri Lanka through a coordinated mobile application. The four members of Group R26-IT-146 investigate related but distinct concerns: foetal monitoring and cardiotocography (CTG) interpretation (IT22272386), maternal-health risk profiling (IT22557506), emotional support with multimodal mood detection (IT22638168), and nutritional assistance (IT22284716). The division of responsibilities is described in the group proposals and the mood-component thesis supplied as contextual evidence. Each individual thesis must account for the author's own contribution rather than claiming ownership of the entire application.

Pregnancy-related dietary needs are part of daily life rather than only clinical appointments. Household food availability, cooking habits, culture, cost and personal preferences constrain which meals can realistically be prepared. Sri Lankan meals may involve locally familiar ingredients such as rice, lentils, vegetables, leafy greens, coconut, fish, eggs and spices; their quantities and methods of preparation materially affect nutrient estimates. A useful digital assistant should help a user discover feasible meals from what is actually available, rather than offering idealised recipes that depend on unavailable ingredients.

The original March 2026 proposal for Member 4, titled Virtual Nutritionist & GDM Action Engine, described a larger system intended to address dietary logging, risk assessment and behavioural interventions. During development, I concentrated primarily on ingredient-based Sri Lankan meal suggestions and nutritional analysis. This thesis deliberately reflects that narrower confirmed scope. It retains the original component name for group traceability but does not describe the broader proposal as though every planned feature had been completed.

1.2 Individual component context

The implemented functional focus is an ingredient-to-meal assistance pathway. The user selects or enters ingredients available at home; the application recommends Sri Lankan meal choices using those ingredients; and the resulting suggestions are accompanied by nutritional information. The approach aims to connect practical meal planning to nutritional awareness in a maternal-health context.

This feature corresponds to the Virtual Chef subsection of the original proposal, which proposed constraining recipe recommendations by available ingredients, nutritional needs or GDM-related constraints, and taste preferences. The proposal mentioned Google Gemini

API and bilingual Sinhala/English recommendations as possible technical mechanisms. These details are preserved as design history, not asserted here as properties of the implemented application because no current code or API evidence was provided. Similarly, nutritional analysis is confirmed as a functional area, but the exact nutrient fields, food database and computation technique require verification from implementation artefacts.

1.3 Problem statement

A pregnant woman deciding what to cook may face three connected problems. First, common recipes may be suggested without considering ingredients already available, leading to impractical recommendations and additional shopping. Second, generic food databases may not represent familiar Sri Lankan dishes or cooking variations sufficiently well to support confident nutritional comparison. Third, recommendations may omit dietary information or imply a degree of nutritional precision that the underlying ingredients, portion estimates or food-composition data cannot justify.

The individual research problem is therefore how to design an ingredient-aware, locally relevant meal-recommendation component that gives understandable and appropriately qualified nutritional analysis. The problem is not to diagnose gestational diabetes or prescribe a meal plan. A nutrition-support component should encourage informed discussion and normal food planning while respecting clinical instructions for users with GDM, allergies or other restrictions.

1.4 Research gap

Prior work cited in the March proposal covers automated food records, nutrition data, glycaemic interpretation and behavioural health applications [3]–[7]. These approaches demonstrate the general feasibility of technology-assisted dietary support but do not by themselves establish how an ingredient-based system performs with mixed Sri Lankan dishes. The gap addressed here is narrower than the original proposal: translating a household ingredient list into feasible Sri Lankan meal alternatives together with transparent nutritional estimates.

A meaningful contribution requires evaluation of both sides of that relation. Recommendation quality must be assessed for local relevance, feasibility, ingredient adherence, dietary constraints and variety. Nutrition analysis must be examined for completeness, source traceability, portion dependence and agreement with a defined reference calculation. The supplied project materials do not include a systematic evidence

review establishing uniqueness against all commercial food applications, so any claim that this is the first system of its kind would be unsupported.

1.5 Aim and objectives

The revised aim is to describe and evaluate an ingredient-based Sri Lankan meal recommendation and nutritional analysis component within the Maathrucare application. The original project aimed to build a much broader Virtual Nutritionist. The objectives below are reformulated to align the individual thesis with the work completed for this component.

- RO1. Define a user-friendly process for selecting and editing household ingredients relevant to Sri Lankan meal preparation.
- RO2. Produce Sri Lankan meal suggestions that use selected ingredients and communicate any additional ingredients or substitutions transparently.
- RO3. Associate suggested meals with understandable nutritional estimates and their necessary assumptions about portions and preparation.
- RO4. Specify and, where evidence permits, assess usability, recommendation relevance, ingredient feasibility and nutritional agreement against a defined reference.
- RO5. Explain how this narrower nutrition component fits within the four-part maternal-health application while keeping clinical decision-making outside its demonstrated scope.

These are revised thesis objectives, not a claim that every evaluation step has already been completed. Section 9.2 links each objective to the evidence currently available and the additional records required. The original proposal's seven objectives are reconciled separately in Table 1.1.

Table 1.1: Scope reconciliation between proposal and implemented component scope

Original proposal area	Status in revised thesis	Reason
Meal/recipe suggestions	Candidate-confirmed implemented focus	Ingredient selection, Sri Lankan suggestions and nutrition analysis
Image recognition / YOLOv8	Proposed; no evidence supplied	Not part of user-confirmed main scope
Sinhala/English voice food logging	Proposed; no evidence supplied	Not verified
OCR and collaborative food learning	Proposed; no evidence supplied	Not verified
Pre-meal GDM risk scoring	Proposed; not claimed implemented	Needs separate clinical design and validation
Hydration, activity and gamification	Proposed; not claimed implemented	No implementation/test artefacts supplied

1.6 Research questions or hypotheses

RQ1: To what degree can users obtain Sri Lankan meal suggestions that are feasible using their selected ingredients? RQ2: How relevant and varied are the suggestions for typical Sri Lankan household scenarios? RQ3: How accurate and interpretable are the nutritional estimates when compared against a specified reference food-composition source and fixed serving assumptions? RQ4: Can users successfully complete the ingredient-selection, recommendation-review and nutritional-comparison workflow without undue effort? RQ5: What design limitations prevent the component from being interpreted as validated clinical GDM advice?

Potential hypotheses for a subsequent evaluation include increased ingredient feasibility compared with non-personalised recipe suggestions and lower task completion time compared with manual recipe lookup. No quantitative hypothesis is marked accepted, because the uploaded documents contain no completed controlled comparison or outcome dataset.

1.7 Scope, limitations and assumptions

My individual contribution focuses on meal suggestions, ingredient selection and nutritional analysis. I describe those functions as my implementation scope and does not infer unseen training pipelines, databases, frameworks, test coverage or deployments. The broader project may offer user profiles and other health information, but actual cross-module integration is not claimed without API traces or a demonstration.

Nutrition estimates depend on identified ingredients, their preparation, serving size and composition references. Different recipes can share a common name but differ substantially in oil, coconut milk, sugar, salt and total yield. Without measured ingredients and a verified reference, an estimate cannot be equated with a laboratory nutrient analysis. The information is for general meal planning, not automatic determination of carbohydrate limits, medicine, insulin needs, foetal health or exercise safety.

1.8 Individual contribution

Nuwantha P.A.T. (IT22284716) has confirmed primary development of three linked functions: choosing ingredients available to the household, generating or retrieving Sri Lankan meal suggestions based on those selections, and presenting nutritional analysis associated with the suggestions. These functions define the individual contribution discussed in Chapters 3–7. Their exact technical implementation and quantitative quality remain subject to review of the codebase and testing records.

The individual does not claim to have developed the teammate's CTG analysis, maternal-risk model or mood-classification models. Nor does this thesis claim that the originally proposed YOLOv8 detector, package-label OCR, blood-glucose-driven risk engine, adaptive pedometer module or gamification system was completed.

1.9 Thesis structure

Chapter 2 reviews relevant literature and the limitations of existing tools. Chapter 3 provides the design-science methodology and proposed evaluation procedures. Chapter 4 describes the implemented functional workflow and the minimum architectural decisions that should be established from the code. Chapter 5 records documented implementation and separates it from unverified technical assumptions. Chapter 6 presents an evidence register and a framework for measured validation. Chapter 7 interprets the contribution and limits. Chapter 8 considers sustainability in Sri Lanka; Chapter 9 summarises achievements and future directions. References and appendices follow.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This working literature review draws on the research areas and bibliography in the my March 2026 proposal, rather than claiming a new systematic search. The most directly relevant themes are technology-assisted food recording, food recognition, recipe recommendation, glycaemic concepts and user interaction. A final review should separately search IEEE Xplore, ACM Digital Library, PubMed and Google Scholar for ingredient-based recipe recommendation, Sri Lankan food-composition studies, recipe constraint satisfaction and accuracy of consumer nutrition estimates. Search dates, inclusion criteria and excluded papers must be documented if a formal review claim is made.

The present review makes an important distinction between established methods and demonstrated results for this specific application. A paper showing promising nutrition assistance in one population is not evidence that the Maathrucare recipe suggestions are safe, accurate or usable. Likewise, prior image-recognition performance is background context, not a result of this implementation.

2.2 Domain and theoretical foundations

Ingredient-aware meal planning can be represented as a constrained recommendation problem. An input set I contains ingredients the user reports as available; a student recipe r

has required ingredients R(r), optional ingredients O(r), serving assumptions and a preparation method. A simple coverage measure is $|I \cap R(r)| / |R(r)|$, although direct matching must be normalised for spelling, synonyms, units and local food names. A recommendation must also disclose which required items are missing and whether substitutions are reasonable. Such a coverage expression is presented as an evaluation definition, not as a claim about the algorithm coded by the student.

Nutritional estimation can be represented as a sum of food-component contributions. For an ingredient i with mass m_i grams and nutrient density n_i per 100 grams, its contribution is $(m_i / 100) \times n_i$; recipe totals are then adjusted for number of servings. This is a general calculation framework. Accuracy also depends on edible portions, cooked/raw composition, cooking losses, recipe yield and source-specific variation. For a mixed Sri Lankan meal, a numeric output without these assumptions may communicate false precision.

Pregnancy introduces particular nutritional considerations, while GDM may require professional guidance regarding carbohydrate intake and glucose monitoring. Glycaemic index and glycaemic load are useful concepts in literature [6], [7], but a recipe nutrient estimate is not automatically a validated predictor of an individual's post-meal glucose response. This distinction is essential for responsible product description.

2.3 Related research

The proposal cites Meyers and colleagues' Im2Calories work [3] and work by Ciocca and colleagues [4] on food images. These studies address food identification and nutrition-support challenges, including the distinction between recognising a food and measuring its consumed quantity. They are relevant context even if the my delivered feature is ingredient selection rather than camera-based recognition.

The proposal also discusses mobile gestational-diabetes reporting [2], glycaemic-index literature [6], [7], and gamification research [8]. Those studies motivated the larger proposed Virtual Nutritionist but should not be presented as components used or evaluated in the narrower meal-suggestion implementation. The research emphasis here instead concerns constrained recipe selection, locally meaningful food terminology and the traceability of nutritional output. A final submission should add verified primary studies specifically about those three areas.

2.4 Existing systems, products and techniques

General-purpose food diaries facilitate meal logging and nutritional summaries, but often depend on searchable databases or manually entered recipes. Standard recipe websites can

suggest meals, yet may not optimise for the ingredients a user currently has or for Sri Lankan food categories. Generative conversational systems can produce recipes quickly, but their ingredient lists and nutritional quantities still require verification. A constraint-based recipe catalogue, in contrast, can return reproducible results if its data are maintained, though it may offer less variety.

These descriptions are conceptual comparisons rather than a recent, exhaustive assessment of commercial products. No evaluated competitor benchmark was provided in the source documents. Consequently, the thesis avoids declaring this application superior in accuracy, coverage or recommendation acceptance.

2.5 Gap synthesis and conceptual framework

The revised research gap lies at the intersection of kitchen feasibility, cultural relevance and understandable nutrition information. Ingredient-only matching would not be enough if suggestions rely on unrealistic quantities or unfamiliar meal patterns. Nutritional calculation alone would not help if the recipe cannot be prepared. The conceptual contribution is therefore a workflow that ties these stages together and allows a user to inspect a recommendation before accepting it.

The evaluation needs three independent judgements: whether a suggested meal is feasible from selected ingredients; whether the recipe is culturally and practically appropriate; and whether the nutrient numbers have a defensible data source and portion basis. These criteria organise the methodology and the eventual results. They do not pre-judge whether the implementation satisfies them.

3 RESEARCH METHODOLOGY

3.1 Research design

The project follows a Design Science Research (DSR) perspective, also identified in the my original proposal. The artefact for this revised study is the ingredient-based meal-planning and nutritional-information component, not the proposed all-feature GDM decision-support ecosystem. Its relevance comes from a practical task: users need meal ideas that match foods in their household and information to compare those choices.

A rigorous DSR account separates problem awareness, construction of a working artefact, demonstration in representative scenarios, and evaluation against stated measures. I developed the central meal assistance functions. However, specific development iterations, dated commits, baseline comparisons and recorded evaluation outcomes have not been

supplied. They must be gathered before claiming an experimental or user-tested research contribution.

3.2 Data, participants or study materials

A recipe evaluation would require a documented set of Sri Lankan meals, ingredient vocabularies, portion assumptions and reference nutrient values. The source materials do not identify a definitive implemented recipe dataset, its number of items, its provenance or its licence. The study cannot therefore report dataset statistics at present. A reproducible version should list each recipe identifier, title, ingredient quantities and units, serving yield, cooking method, source and last update.

Representative task scenarios should include common combinations of household staples and their variations, such as meals based on rice, lentils, vegetables, leafy greens and selected protein sources. Scenarios should also cover sparse ingredient sets, misspellings, duplicate ingredient names, substitutions and zero-result cases. Such examples are proposed test designs, not a statement that these records already exist in a database.

No ethically approved participant study, dietitian review or user response dataset has been provided. Accordingly, this draft does not claim that pregnant women were recruited, surveyed or assessed, nor that clinical professionals approved the produced nutritional advice.

3.3 Experimental or study procedure

A reproducible study should begin with a frozen version of the code, recipe data and nutrient-reference tables. First, I would document the current ingredient-selection UI and save a controlled set of input ingredient lists. Second, the implemented recommendation endpoint or function would be run on those lists, recording the students returned, their displayed ingredient requirements and any substitutions. Third, blinded reviewers familiar with Sri Lankan cooking would score the outputs for feasibility and cultural relevance. Fourth, the recipe nutrition estimates would be compared with independent calculations made from the same quantities, edible portions and reference table. Finally, a usability exercise could record task success, completion time, corrections and user comprehension.

The sequence is deliberately modular. If the application is not accessible, evaluation can still begin from versioned sample inputs and captured outputs. A prototype screenshot can demonstrate a user interface but does not by itself establish the correctness of recommendations or nutrient totals. The final report should identify who ran each test, on what date, under which code revision and with what evidence.

3.4 Measures and evaluation metrics

Recommendation relevance and nutrient correctness should not be combined into a single vague accuracy percentage. Relevance measures the usefulness of suggested meals for the selected ingredients. Nutrition correctness measures agreement with a defined numerical reference and relies on fixed recipe quantities. Usability evaluates whether users can operate the workflow. Safety checks assess whether claims and constraints are handled appropriately.

A student acceptance target should be set before the evaluator inspects final outcomes; selecting a convenient target after results are known would risk hindsight bias. The original proposal's targets for YOLOv8 recognition and GDM risk prediction are not appropriate substitutes for metrics of ingredient-based recipe suggestions.

Table 3.1: Evaluation measures and required evidence

Measure	Definition / protocol	Required evidence
Ingredient feasibility	Proportion of suggestions whose mandatory ingredients are available, or whose missing items are clearly disclosed	Frozen inputs, recipe output and independent item audit
Top-k relevance	Reviewer rating for a fixed number of suggested recipes on local suitability and usefulness	Review rubric, multiple raters and raw ratings
Variety / diversity	Unique appropriate recommendations across repeated or varied ingredient sets	Batch outputs and deduplication rules
Nutrition absolute error	Absolute difference per nutrient versus a specified reference calculation for the same serving	Reference data, ingredient masses and computed totals
Nutrition completeness	Share of displayed nutrient fields with a valid source and clearly stated units	Data lineage and screen audit
Task success / time	Completion of select -> suggest -> inspect sequence and elapsed seconds	Observed sessions or recorded usability runs
Constraint failures	Allergy, excluded ingredient, missing portion, unsupported food and unsafe statement cases	Negative tests with documented expected outcomes

3.5 Data analysis

For numeric measurements, the final analysis should report sample size, central tendency and variability, along with per-category patterns. Nutrition errors should be broken down by nutrient and by recipe instead of reporting only a pooled average. Recommendation ratings

should present distributions, rater agreement where more than one judge is used, and examples of failed recommendations. Usability results should include the proportion of tasks completed without assistance and distinguish user confusion from service errors.

An ingredient feasibility metric can be misleading if every system suggestion is highly repetitive; diversity must therefore be examined separately. Nutrition agreement can also be misleading when both the system and benchmark use the same incorrect portion assumption. Whenever possible, a separately prepared benchmark should be used, with the inputs and limitations documented. If a sample is small, descriptive findings should take precedence over unsupported statistical generalisation.

3.6 Ethics, privacy, security and responsible AI

Pregnancy and GDM are health-related contexts. The component should make a clear separation between food information and personalised medical advice. It must not present a meal recommendation as a guarantee of safe blood glucose, nor infer a diagnosis from an estimated nutrient table. Any diet restrictions, allergies or clinician-prescribed plans require careful handling; where the interface cannot reliably interpret a restriction, it should defer rather than offer an assertive recommendation.

Ingredient lists may appear harmless, but user profiles or linked health records can disclose sensitive information. Any user study requires appropriate information, consent and data minimisation. If an external generative service is used, the actual terms, transmitted fields, model behaviour and retention settings should be checked before describing data as private or secure. Prompt-generated nutritional figures need an independently maintained reference source and validation before being shown as factual.

3.7 Reproducibility and research artefacts

The minimum reproducibility package should contain a repository revision identifier; language/framework versions; a description of the recommendation technique (rule-based, search, model-assisted or hybrid); permitted ingredient vocabulary; recipe data or anonymised test fixtures; nutritional reference source; representative request/response examples; screenshots; and scripts that can rerun evaluation. Test data used for choosing algorithms must be separated from held-out evaluation scenarios to avoid overfitting the final report to examples already debugged.

The supplied artefacts now include validation screenshots, model checkpoints and source-code inspection notes, but a full repository snapshot, training manifest, complete database

export and controlled benchmark records remain unavailable. Appendix A lists the evidence supplied and the remaining reproducibility requirements.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The revised requirements prioritise the user journey that the I developed. Essential tasks are identifying ingredients, requesting meal suggestions, reviewing locally relevant results and inspecting nutrition information. Quality attributes include comprehensibility, graceful handling of empty results, consistency of nutrient units and the ability to explain whether additional ingredients are needed. Advanced disease-specific suggestions are deferred until validation and appropriate clinical constraints are available.

Table 4.1: Functional requirements and current evidence state

ID	Requirement	Evidence status
FR-01	User can select or edit a set of available ingredients	Candidate-confirmed function; screenshot/code needed
FR-02	System provides Sri Lankan meal suggestions informed by selections	Candidate-confirmed function; outputs needed
FR-03	Meal suggestions include nutritional analysis or summaries	Candidate-confirmed function; sources/fields needed
FR-04	Display ingredient requirements and substitutions transparently	Design/evaluation requirement; implementation unverified
FR-05	Support quantities/portions and correct nutrient units	Design/evaluation requirement; implementation unverified
FR-06	Handle no matching recipes or insufficient information safely	Required verification; implementation unverified
NFR-01	Readable mobile layout and understandable labels	Usability evidence pending
NFR-02	Protect user data and restrict access appropriately	Implementation/security evidence pending

4.2 Component architecture within the complete solution

Maathrucare's group architecture includes four separate contribution areas. The nutrition module supports the meal decision process and can coexist with CTG, risk-monitoring and mood-support functions. The teammate's October 2026 draft explicitly identifies IT22284716 as providing meal suggestions from ingredients available at home, with nutritional analysis. That cross-check is consistent with the my latest confirmation of actual scope.

The conceptual diagram below depicts ownership, not a verified set of network connections. No specific endpoint or shared database contract has been supplied. Therefore, arrows should not be interpreted as proof that all four components exchange patient data in production.

MAATHRUCARE pregnancy-support application (group ecosystem)
↓
Parallel research areas: CTG | Maternal risk | Mood chatbot | Nutrition
↓
FOCUS OF THIS THESIS: selected ingredients -> Sri Lankan meals -> nutritional analysis
↓
User-facing nutrition information (not clinical decision-making)

Figure 4.1: Nutrition component within Maathrucare

4.3 Design alternatives and decisions

Ingredient-based suggestion systems can be built through exact catalogue matching, flexible rule-based scoring, similarity retrieval, prompt-based recipe generation or a hybrid pipeline. These techniques have different trade-offs. Fixed recipes are easier to reproduce and audit; generated recipes may offer greater variety but can invent quantities, ingredients or nutrient values; ranking systems can promote feasibility without hard-filtering out every recipe with one absent seasoning. The original proposal anticipated Gemini-assisted Virtual Chef outputs, but the actual choice in the application must be verified from the repository.

Nutrition analysis may be read from precomputed recipe records or calculated from ingredient masses. Precomputed values are inexpensive at runtime but become wrong when portions or substitutions change. Dynamic calculation is more transparent if ingredient quantities and an authoritative composition reference are recorded, but it must handle cooking losses, ambiguous quantities and local ingredient names. The final thesis should state which method was actually used and show an example trace from raw ingredients to displayed results.

Table 4.2: Design alternatives

Decision area	Option A	Option B	Verification needed
Recommendation	Curated recipe filtering/ranking	Generative or hybrid suggestion	Actual code flow and ranking rules
Ingredients	Fixed selection vocabulary	Free-text entry + normalisation	Stored values and synonym handling
Nutrition	Recipe-level stored totals	Ingredient-based calculated totals	Data source, units and method

Decision area	Option A	Option B	Verification needed
Local language	English-only labels	Sinhala/English interface	Actual UI screenshots / localisations
Safety	General information only	Clinical-rule constraints	Claim review and any clinician approvals

4.4 Detailed design

The minimum confirmed interaction begins when a user specifies available ingredients. The system then proposes one or more Sri Lankan meals derived from those choices and provides corresponding nutrition details. A robust design should retain the exact input set used for each recommendation so a reviewer can explain why a meal appeared. The recipe result should distinguish ingredients already selected from additional ingredients, and it should describe its portion assumption. The user should be able to correct selections and request another suggestion if the proposed recipe does not match their household circumstances.

The data fields below illustrate the input/output fields needed for a reproducible evaluation. The implementation has also been described at the table level in Chapter 5; these remain a recommended research contract, not proof of an exact REST response schema.

Table 4.3: Proposed minimum recipe data contract

Field	Purpose	Example / rule
input_ingredients	Reproducible input to recommendation	Normalised IDs, not only display labels
recipe_id, recipe_name	Stable identifier and local meal name	One record per suggested meal
used / missing ingredients	Feasibility explanation	Keep optional ingredients separate
quantities, servings	Basis for estimation	Document grams and yield
nutrients_per_serving	Calories, carbohydrate, protein, etc. if available	Units and rounding explicit
nutrient_source, version	Traceability	Catalogue or reference citation
restrictions_checked	Safety / exclusion handling	Never assume medical safety from absence of flag

User selects ingredients available in the household
↓
Input confirmation and normalisation (implementation detail to verify)
↓
Sri Lankan meal recommendation / selection (candidate-confirmed function)
↓

Display recipe information and nutritional analysis (candidate-confirmed function)
↓
User reviews, compares and revises ingredients as needed

Figure 4.2: Ingredient-to-meal recommendation and nutrition review workflow

The workflow illustration separates confirmed user-visible functionality from unverified internal stages. Input normalisation, ranking rules, feedback capture and iterative correction may have been implemented, but they are not claimed here without inspection. A complete design account should include at least two true screenshots: the ingredient-selection page and one suggestion with its nutritional information, preferably accompanied by the input that produced it.

Nutrition calculation should be interpreted cautiously. If nutrient densities are known per 100 g and ingredient quantities are fixed, per-serving totals can be computed by summing ingredient contributions and dividing by the declared number of servings. This well-understood formula can form a reference test independent of the app. It does not establish that the current implementation uses this approach; any generative estimates need their own validation.

4.5 Standards and professional practice

The development and reporting process should maintain identifiable requirements, testable data contracts, error handling, accessibility consideration and source-controlled changes. Responsible nutrition reporting calls for units, reference data, portion assumptions and understandable uncertainty language. Clinical claims require an additional level of professional review that is not established by this project documentation. Where recipes have uncertain quantities or incomplete food-composition values, the interface should avoid unwarranted numerical precision.

If the application stores health information, access control and secure transport should be independently tested. A user must understand what information is derived from selected ingredients and what, if anything, is obtained from another Maathrucare component. The thesis should not claim compliance certification, regulatory approval or clinical validation unless independently evidenced.

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development environment and technology stack

The inspected technical walkthrough identifies React Native 0.86.3, Expo SDK 57, Expo Router, NativeWind, Expo Image Manipulator and Expo File System in the Maathru Care client. It reports Supabase PostgreSQL for users, maternal_foods, meal_logs and meal_items; a Python FastAPI image-inference service hosted on Render; and a custom YOLOv8 detector. These details are taken from a user-supplied source-code inspection transcript (Appendix C), not from an independently mounted repository. This is an Expo/React Native application with a possible web target, and should not be characterised as a web-only application. The March 2026 proposal anticipated Flutter/Firebase; the later walkthrough indicates a changed implementation stack.

Key client implementation files named in the supplied code-inspection report are src/app/(mother)/meal-scan.tsx, src/app/(mother)/gdm-care.tsx, src/services/api/foodDetector.ts, src/services/mealService.ts, src/types/database.types.ts, src/types/meal.ts, and src/constants/pregnancyNutritionTargets.ts. Version claims, database fields, endpoints and deployment status must be checked against the exact submission commit and backend repository before final endorsement.

5.2 Component implementation

The implemented nutrition workflow has expanded beyond the earlier narrow meal-suggestion description. In addition to ingredient-linked Sri Lankan meal suggestions and nutritional summaries, the new developer-supplied code-inspection transcript describes image-based meal scanning through a custom YOLOv8 inference API. The client selects a camera image, resizes it to a width of 640 pixels, compresses it as JPEG (quality 0.7), and sends it to the inference service. Detected food labels and confidence values are returned for user review; detections exceeding a reported 0.50 confidence cutoff are automatically selected. Ingredient selection and meal suggestions remain part of the individually reported contribution, but their exact source functions need direct repository confirmation.

The meal service maps model class labels to database food entries. Reported normalisation trims and lowercases identifiers, replaces spaces/hyphens with underscores, and removes unsupported characters. An exact maternal_foods.food_id lookup is attempted first, followed by a case-insensitive display_name fallback. The selected foods can form a meal record, and portion multipliers adjust nutrient amounts before recording. This is a documented client

logic design, not a measured accuracy result; false matches from the fallback require explicit tests.

For each logged meal the documented relational flow is users (profile) → meal_logs (per-user meal and day totals) → meal_items (individual food items and serving_multiplier) → maternal_foods (food composition and glycaemic-index category). Meal-specific nutritional fields include carbohydrate, sugar, fibre, fat, protein (in reference records), iron and calcium. The submitted walkthrough also describes warning logic before saving a meal: for a reported high-risk GDM state, sugar above 8 g per meal or carbohydrate above 50 g per meal raises warnings; projected daily sugar and carbohydrate totals also trigger warnings. These software thresholds require clinician review: they are implementation rules, not established diagnostic cut-offs or evidence of medical safety.

Table 5.1: Confirmed scope and verification status

Function	Current statement	Evidence to attach
Ingredient selection	Implemented according to candidate	Screen or recording, state handling, exact input
Sri Lankan meal suggestions	Implemented according to candidate and teammate summary	Example outputs, recipe source, algorithm or prompt
Nutritional analysis	Implemented according to candidate and teammate summary	Nutrient fields, source and calculation trace
Full GDM risk-score engine	Proposed originally; not claimed implemented	Only include if separately verified
YOLOv8 / OCR / voice capture	Proposed originally; not claimed implemented	Do not include as completed without evidence
Exercise / hydration / gamification	Proposed originally; not claimed implemented	Leave in future work unless demonstrated

5.3 Integration with other components

The nutrition module sits within the shared Maathrucare product. The CTG component concerns fetal monitoring, the maternal-risk component concerns patient health indicators and doctor notification, and the chatbot component concerns mood and supportive conversation. Nothing in the supplied material demonstrates a tested end-to-end nutritional data interface between the my module and all three others. It is sufficient to describe these interfaces as potential integration points until an implemented API is shown.

Where integration does exist, the thesis should document message fields, optional values, freshness, responsibility for consent and the component that owns clinical decisions. A nutrient estimate should not be translated automatically into an alert or GDM classification

without explicit design and verification. The strongest defensible present integration claim is that the meal recommendation feature belongs to the same group pregnancy-support application.

5.4 Verification and software testing

The supplied code-inspection transcript is a useful documentary source of implementation logic but is not a substitute for executable unit tests, API traces, database evidence or model-validation artefacts. Functional verification should exercise: valid image upload; low-confidence food results; no food detected; label-to-food matching; missing Supabase food rows; portion-multiplier arithmetic; transaction failure while writing logs/items; GDM warning paths; and camera/network permission errors. A controlled testing report should record test ID, code revision, fixture, expected output, observed output, pass/fail, and timestamp. Do not mark a scenario passed without a saved run.

Tests should stress whether the recommendations really depend on selected inputs. A generic dish returned for every combination is not equivalent to ingredient-aware recommendation. Nutritional tests should assess portion change, ingredient replacement, unit conversions and missing data. Safety-oriented negative tests should include an excluded ingredient, an unknown ingredient and ambiguous recipe quantities.

Table 5.2: Recommended functional test scenarios

Test ID	Input / task	Acceptance observation	Status
TC-01	Choose a realistic household ingredient set	Recommendations explain ingredient fit	Not evidenced
TC-02	Remove a key ingredient and regenerate	Output or missing-item indication changes appropriately	Not evidenced
TC-03	Choose no ingredients	Helpful empty-state behaviour	Not evidenced
TC-04	Unknown or misspelled ingredient	Predictable correction/fallback	Not evidenced
TC-05	Inspect nutrient values for one recipe	All shown units and portion basis are understandable	Not evidenced
TC-06	Change serving number or recipe quantity	Nutrients scale or limitations are explained	Not evidenced
TC-07	Try an excluded/allergen ingredient	No false assurance of safe suitability	Not evidenced
TC-08	Network/API failure if applicable	Non-misleading error and retry path	Not evidenced

5.5 Deployment and operation

The walkthrough specifies an image-inference endpoint on a Render-hosted FastAPI service and a Supabase-hosted PostgreSQL backend. This documents the intended/observed service configuration in the inspected code, not independently verified service uptime, access control, clinical suitability or production availability. Deployment evidence to retain includes redacted configuration, API health checks, database policies, request logs and an authenticated demo capture. Secrets and patient-identifiable data must not be included in appendices.

6 RESULTS AND VALIDATION

6.1 Research results

Figure 6.1 presents the saved YOLO validation output supplied with my project materials. The custom YOLO food detector reports precision = 0.9432 (94.32%), recall = 0.9417 (94.17%), mAP@50 = 0.9392 (93.92%), and mAP@50–95 = 0.9298 (92.98%). The same run excerpt reports 1.2 ms preprocessing, 3.7 ms inference, and 1.3 ms postprocessing per image in its validation environment; these are not mobile-device or network end-to-end timings. The screenshot gives /content/runs/detect/val-3 as the output location. Its dataset size, splits, input hardware, evaluation command, confidence and IoU settings cannot be independently determined from the partial screenshot. I report these metrics from the saved validation screenshot. I have not rerun the model validation in preparing this document, so I distinguish the recorded output from a new independent experiment.

Figure 6.2 documents the supplied class-normalised confusion matrix. The visible labelled categories include asmi, chicken_bun, chicken_cutlet, vegetable_roll, vegetable_roti, vegetable_pastry, pizza variants, kokis, konda_kavum, mun_kavum, soft drinks and milkshake. Most foreground-class diagonal cells are between 0.97 and 1.00, but the background column shows substantial assignments to vegetable_roll; 0.89 appears at the vegetable_roll row against the background column. Thus the matrix does not justify a claim of near-perfect behaviour in all conditions. Figure 6.3 shows predictions on a selection of images; multiple boxes on the same item and crowded overlapping labels are visible in some examples. These illustrate potential duplicate detections and the importance of per-object review and suppression, despite high aggregate validation metrics. The underlying image count, class sample sizes and image identities have not been supplied.

Table 6.1: Evidence and results register

Outcome	Available evidence	Present conclusion / next step
Ingredient-selection function	Student confirmation	Implemented scope reported; add demo and test
Sri Lankan meal suggestions	Student confirmation; teammate description	Implemented scope reported; audit true outputs
Nutritional information display	Student confirmation; teammate description	Implemented scope reported; reference/calculation unknown
Ingredient coverage / relevance	No review dataset	Not evaluated
Nutrient estimation error	No reference comparison	Not evaluated
User usability / satisfaction	No participant results	Not evaluated
Improved GDM outcomes	No clinical study	Not evaluated; no clinical benefit claim
Broader proposed AI modules (scope adjusted after code-inspection evidence)	Proposal only	YOLO food detection described in developer report; OCR/voice/exercise not demonstrated
Custom YOLO detector	Reported 0.9392 mAP@50	Student-provided value; original training logs pending

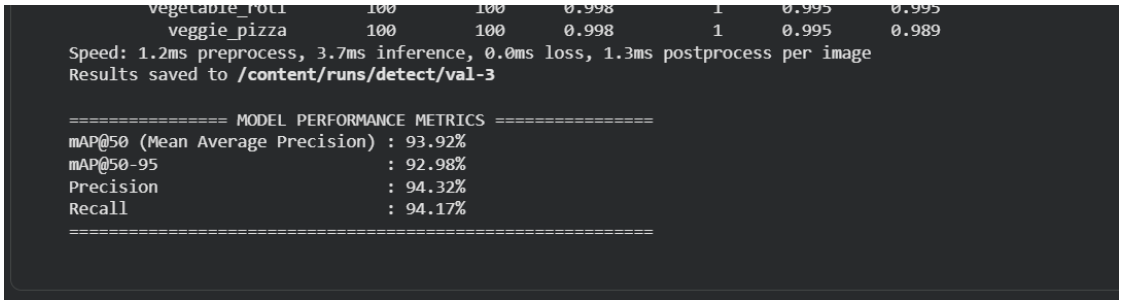


Figure 6.1: Recorded YOLO validation run summary (precision, recall, mAP@50 and mAP@50–95).

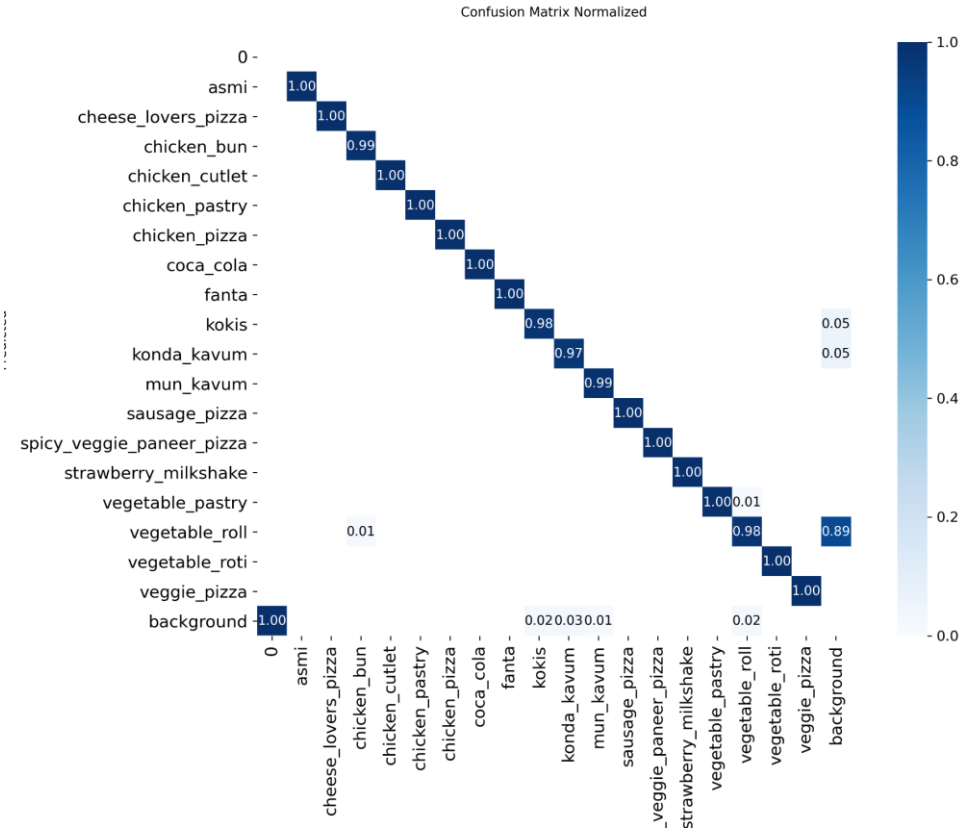


Figure 6.2: Normalized class confusion matrix from the supplied YOLO validation artefacts.



Figure 6.3: Food-object detector sample predictions, showing localized boxes and some crowded or overlapping predictions.

6.2 Product and technical results

The source-code inspection identifies a complete intended chain: mobile camera/image picker → resize/compress → YOLO FastAPI inference → confidence review → food-id normalization → maternal_foods nutrient lookup → per-portion nutrient display → meal_logs and meal_items persistence. Supabase stores a clinical user profile, food records, meal-level summaries and component-level records. The client additionally displays GDM-related food warnings calculated from current intake totals. These are code-documented functions; claims of successful end-to-end performance require test-run evidence.

The YOLO validation screenshot reports mAP@50 93.92%, mAP@50–95 92.98%, precision 94.32%, and recall 94.17%. These are object-detection results and cannot be equated with ingredient-based recipe relevance, the precision of nutrient values, GDM outcome

improvement, or diagnostic accuracy. The provided validation montage and confusion matrix also show repeated boxes for some objects and a notable vegetable_roll/background ambiguity. FAO Sri Lanka Food Composition Table (2021) and USDA FoodData Central are identified as independent reference sources for nutrient-value comparison, not verified imports into the current maternal_foods table. The Kaggle GDM dataset is separate clinical research material and has not been shown to validate the YOLO model or the maternal-food composition database.

6.3 User or stakeholder validation

The proposal anticipated a participant study and dietitian review, but no participant responses, sample description, consent records or expert scoring sheets have been provided. This thesis consequently makes no assertion that pregnant women preferred the suggestions, that the system improved dietary adherence or that clinicians approved any nutritional values. A future study can test meal suitability with local home cooks and nutrition professionals, and can separately measure usability with representative users. The participant characteristics and any ethical approvals must be recorded before results are published.

6.4 Integration results

The reviewed application architecture connects the nutrition client to a food-detection service and Supabase; those connections are within the nutrition workflow. Cross-member handoffs to CTG, risk profiling and the mood chatbot should be reported only if an actual interface or shared component is demonstrated. The nutrition module can read a risk category as context for food warnings, but that does not make its heuristic warning thresholds a validated GDM diagnostic model.

6.5 Summary of findings

Measured model evidence now includes four values transcribed from the supplied validation run: precision 0.9432, recall 0.9417, mAP@50 0.9392, and mAP@50–95 0.9298, accompanied by a normalized confusion matrix and qualitative detection examples. The prediction montage reveals some overlapping/duplicate detections, which should be investigated in an independent test set. Two best.pt files were submitted as model checkpoints; they are not assumed identical, nor was one selected as the precise source of the displayed metrics without matching validation metadata. End-to-end food/nutrition logging, nutrition data accuracy and actual maternal-health outcomes require separate tests. The revised conclusion is limited to the model evidence and the implementation characteristics corroborated by the supplied documentation.

7 DISCUSSION

7.1 Interpretation of findings

An ingredient-based approach has practical relevance because it begins with the resources available to a household. This differs from nutrition planning that presents an ideal meal without regard to food availability. Its value depends on correctly matching foods to user inputs and presenting nutrition in a form that does not confuse estimates with guarantees. The my reported implemented functions provide a basis for investigating these questions, but the current materials cannot establish their performance.

The apparent narrowing from a seven-objective Virtual Nutritionist proposal to a three-function meal-planning implementation is not inherently a defect. Research and development often require prioritisation. Academic quality instead depends on describing the actual work precisely, explaining the design decisions and testing it appropriately. Overstating unfinished features would make the report less trustworthy even if the functioning meal suggestion feature itself is useful.

7.2 Comparison with literature baselines and existing solutions

The early literature on photo-based food diaries [3], [4] addresses recognising what was consumed, whereas the confirmed contribution begins before preparation with ingredients the user already has. That functional difference motivates a different benchmark. A meaningful comparison should use a baseline that offers generic recipe search or non-personalised suggestions and measure how often student recipes can truly be made from selected ingredients. Because no such evaluation has been supplied, this chapter cannot assert improvement over published results or consumer applications.

7.3 Research contribution

The individual research contribution is best framed as a culturally situated application of ingredient-constrained meal recommendation linked to nutrition information. The contribution will be strongest when the I demonstrate the selection/matching logic and submits systematic evidence showing when the recommendations are appropriate and when they fail. It should not be presented as a novel clinical risk algorithm unless one was truly developed and validated.

7.4 Product contribution

My product contribution comprises the ingredient selection interaction, Sri Lankan meal suggestion outputs and their nutritional analysis presentation. These elements form one recognisable workflow for a pregnant user planning a meal at home. Their exact source files, data tables and build environment should be documented in the submission appendix to make individual ownership verifiable. Other team members retain their respective research modules.

7.5 Limitations and threats to validity

Internal validity is limited by the absence of recorded test runs and a declared recommendation baseline. Construct validity would be threatened by labelling a dish 'feasible' when a missing ingredient is essential, or calling a nutrient prediction accurate without checking portion assumptions. External validity is limited by the diversity of Sri Lankan recipes, regional terminology, household budgets, dietary practices and mobile-device access. Dataset selection bias is possible if evaluation uses only recipes already built into the system.

Nutritional completeness is a separate concern from correctness: a summary may display energy and protein while omitting sodium, fibre or micronutrients. Conversely, an extensive table of nutrient values may be unhelpful if based on approximations not explained to users. Clinical validity has not been established; the component must not be interpreted as proof of safe GDM meal selection or a replacement for professional care.

7.6 Implications

For users, the design highlights a practical route from available kitchen ingredients to locally recognisable meal ideas. For researchers, it suggests evaluation measures distinct from conventional food classification accuracy. For developers of the group application, it clarifies the contract between general nutritional assistance and the separate maternal-health risk module. Any future move towards disease-specific decision support should be governed by clinical review, auditability and explicit safety limits.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The immediate value proposition is reducing the work of finding practical Sri Lankan meal ideas while helping users compare their nutritional content. Pregnant women and family

members who prepare meals are the primary intended users. Secondary beneficiaries may include clinicians who provide general dietary education, but the current module is not an approved dietary treatment system and must not be marketed as such.

8.2 Market and competitive position

The concept may be differentiated by its Sri Lankan food terminology and its focus on household ingredient availability. These are product positioning hypotheses, not proven market advantages. A future competitive assessment should compare actual local applications and recipe platforms on language support, ability to handle regional dishes, ingredient feasibility and traceability of nutrient values. Market-share figures and willingness-to-pay estimates cannot be inferred from the project proposals.

8.3 Business or sustainability model

Potential sustainability pathways include a free educational meal-idea tool supported by a curated recipe database, or a supported institutional deployment within a health-education programme. Subscription or partnerships would require user research, a maintenance budget and clarity about what services are medically reviewed. The recurring cost depends on the actual implementation: catalogue search has different infrastructure requirements from generative inference and continuous real-time biometrics.

Table 8.1: Sustainability considerations

Issue	Practical implication	Necessary action
Sri Lankan recipe coverage	Varied ingredient names and preparation methods	Curate and version local meal records
Nutrient data quality	User trust and potential safety concerns	Attach verifiable references and portion assumptions
Generative API dependency if used	Cost, failures and changing outputs	Cache, version, review, provide fallback
Accessibility and language	Potential exclusion of intended users	Test on realistic phones with target-language speakers
Clinical claims	Risk of misleading GDM advice	Keep scope general until medically validated

8.4 Scalability, intellectual property and compliance

The main scaling needs are reliable recipe maintenance, transparent nutritional updates, multilingual ingredient names, server capacity if applicable and support for user corrections. Rights to recipe text, images, food data and any model outputs must be checked rather than assumed. Storage and sharing of pregnancy-related personal data require appropriate privacy

and access-control design. No formal compliance or regulatory certification is established by the uploaded materials.

8.5 Contribution to the Sustainable Development Goals

The work is relevant to SDG 3 (Good Health and Well-Being) by exploring access to practical nutrition information, and potentially to SDG 9 (Industry, Innovation and Infrastructure) through locally contextualised software. These are thematic alignments, not evidence that the application improved maternal health outcomes. Demonstrating such impact would require a separate evaluated deployment.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis revises the individual component to match the work actually reported by Nuwantha P.A.T. (IT22284716): an ingredient-based Sri Lankan meal recommendation workflow accompanied by nutritional analysis. It situates that contribution within the Maathrucare research project and distinguishes the delivered functional emphasis from the larger Virtual Nutritionist & GDM Action Engine originally proposed. The implemented feature addresses a practical need, helping users explore meal ideas based on available household foods. Its research value will depend on whether future testing establishes that those recommendations are feasible, culturally appropriate and supported by defensible nutritional information.

The uploaded evidence does not support numerical accuracy, health-benefit, clinician-validation or full system-integration claims. The final document should therefore attach real implementation artefacts and evaluation data before being submitted as a completed, empirically validated dissertation.

9.2 Achievement of objectives

The objectives are mapped below to the present state of knowledge. 'Documented implementation' identifies features described in the technical materials whose complete end-to-end test evidence is not attached; 'pending' means the relevant evaluation evidence is not present.

Table 9.1: Objective achievement register

Objective	Current state	Evidence still required
RO1: Ingredient selection	Documented	Screenshots, input state and code
RO2: Sri Lankan meal suggestions	Documented	Request/response examples and feasibility review
RO3: Nutritional analysis	Documented	Nutrient data source and independent calculation check
RO4: Quality and usability evaluation	Pending	Test dataset, reviewers and summary metrics
RO5: Group context and safe scope	Documented conceptually	Actual integration contracts, if implemented

9.3 Individual research and product contributions

My documented contribution concerns ingredient-aware Sri Lankan meal assistance, a YOLO-linked food scanning pathway, nutritional summaries and related Supabase meal logging. The supplied source-code inspection records the roles of meal-scan.tsx, foodDetector.ts and mealService.ts; attribution of individual commits, backend model training and SQL migrations must be substantiated with repository records. The separate CTG, maternal-risk and mood-classification models are credited to the other group members.

9.4 Future directions

The most urgent next step is empirical validation of the feature already built: freeze the recipe set and matching code, collect authentic recommendation outputs, assemble independent nutrition benchmarks and conduct usability and local relevance reviews. Fixing incorrect ingredient matches and making serving assumptions explicit should take priority over adding more speculative features. Robust handling of allergies, exclusions and ambiguous ingredient quantities is a prerequisite for more personalised advice.

Once the core pathway has been documented and assessed, the broader original proposal may guide future work: computer-vision food recognition, Sinhala/English voice meal logging, OCR, richer nutrient profiling, clinically reviewed GDM decision support, hydration reminders, activity tracking and gamification. These extensions should be introduced as separate work packages with their own evaluation criteria, particularly where maternal health and blood-glucose interpretation are involved.

REFERENCES

The following sources were checked against publisher, official documentation or source landing pages. Online references are used to describe datasets and definitions; they do not

establish that those datasets were imported into the application. Dates accessed: 11 October 2026.

[1] R. Jayatissa, A. G. Perera, A. B. G. Silva, Y. N. Amarathunga, N. D. De Alwis, A. Anathana, and T. Longvah, Sri Lanka Food Composition Table (SLFCT), 2nd ed., Colombo, Sri Lanka, 2021. FAO/INFOODS catalogue: https://www.fao.org/food-composition/tables-and-databases/detail/(sri-lanka--2021)-sri-lanka-food-composition-table/en

[2] U.S. Department of Agriculture, Agricultural Research Service, FoodData Central: Download Datasets, 2026. https://fdc.nal.usda.gov/download-datasets/

[3] Ultralytics, YOLO Performance Metrics, documentation. https://docs.ultralytics.com/guides/yolo-performance-metrics/

[4] Ultralytics, Model Evaluation and Fine-Tuning, documentation. https://docs.ultralytics.com/guides/model-evaluation-insights/

[5] Ultralytics, Ultralytics YOLO GitHub repository and documentation. https://github.com/ultralytics/ultralytics ; https://docs.ultralytics.com/

[6] S. Santhosh, Gestational Diabetes Mellitus (GDM Data Set), Kaggle, dataset landing page. https://www.kaggle.com/datasets/sumathisanthosh/gestational-diabetes-mellitus-gdm-data-set/data . Supplementary student resource; not shown to be the YOLO dataset.

[7] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, “The Pascal Visual Object Classes (VOC) Challenge,” International Journal of Computer Vision, vol. 88, pp. 303–338, 2010. https://doi.org/10.1007/s11263-009-0275-4

[8] U.S. Department of Agriculture, FoodData Central: Data documentation and definitions. https://fdc.nal.usda.gov/

Internal project evidence: Nuwantha P.A.T., Virtual Nutritionist & GDM Action Engine proposal, March 2026; supplied Maathru Care source-code inspection transcript, 11 October 2026; S. D. Livera, IT22638168 individual thesis draft, October 2026. The precise inspection transcript is retained in the project evidence bundle.

GLOSSARY

Ingredient feasibility — The extent to which a meal suggestion can be prepared from the selected household ingredients.

Nutritional analysis — A displayed estimate or calculation of nutrients in a specified meal or serving, dependent on its data source and assumptions.

Portion assumption — The stated quantity or number of servings to which nutrient numbers apply.

Recipe recommendation — The process of returning student meals according to user inputs or preferences.

Clinical validation — A structured professional assessment of a health-related method or recommendation; not established in this thesis.

Documented implementation — Functionality the student confirms was developed, awaiting verification against the source code and observed output.

APPENDICES

Appendix A — Validation and research evidence register

Validation screenshot: precision 0.9432, recall 0.9417, mAP@50 0.9392, mAP@50–95 0.9298. Output directory in provided excerpt: /content/runs/detect/val-3. Also supplied: normalized confusion matrix; food-detection prediction montage; two best.pt model-checkpoint files. These materials provide substantially more evidence than a single undocumented mAP figure, but do not establish the underlying evaluation image count, split provenance, training configuration, model-checkpoint identity or independent deployment quality. The original dataset manifest, training configuration YAML, per-class support and evaluation command remain needed for reproducibility.

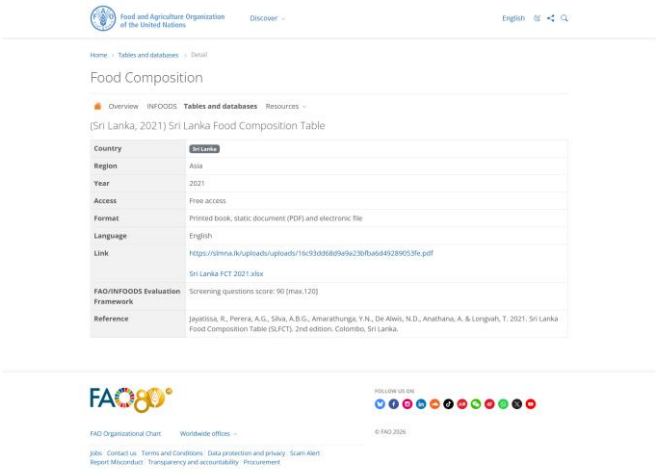
Table A.1: Proposed recipe evaluation form

Review field	Question for each recommendation	Recorded value
Input ingredients	What exact ingredients were selected?	Not submitted; record observed evidence
Suggested recipe	What named Sri Lankan meal was returned?	Not submitted; record observed evidence
Feasibility	Can the dish be made without undisclosed required ingredients?	Not submitted; record observed evidence
Local appropriateness	Would this combination be plausible in the intended context?	Not submitted; record observed evidence
Serving definition	What quantity or serving yield is assumed?	Not submitted; record observed evidence
Nutrient correctness	How does each displayed value compare with the reference?	Not submitted; record observed evidence

Review field	Question for each recommendation	Recorded value
Reviewer / timestamp	Who performed the assessment, and when?	Not submitted; record observed evidence

Table A.2: Research risk register

Risk	Why it matters	Proposed mitigation
Invented or implausible recipes	Poor usability or inappropriate meals	Recipe review and clear missing-ingredient notice
Incorrect nutrient estimates	False impression of dietary suitability	Composition-source audit and independent calculation
Unspecified serving size	Comparisons become misleading	Label quantities and scaling assumptions
Allergen / dietary restriction failure	Potential harm	Negative tests and conservative advice
Unsupported medical claim	Confuses nutrition ideas with GDM treatment	Explicit boundary and qualified clinical review
Unavailable recipe/model service	Missing results in daily use	Controlled fallback and retry
No evaluation evidence	Weak research claims	Versioned tests and authentic results collection



Academic integrity and AI-use disclosure. Generative AI assistance was used to organise and edit this thesis, compare the provided campus template and group-member draft, interpret user-supplied code-inspection documentation, format technical tables, and incorporate research screenshots and evidence references. No synthetic model scores, participant responses, nutritional benchmarks, or clinical-validation outcomes have been introduced as real results. The four YOLO metrics and evaluation figures are reproduced from the student-supplied validation screenshots. The student is responsible for checking every technical statement, accurately disclosing any AI tools used during development and research, confirming final citations, and complying with the applicable SLIIT policy. Supervisor and student signatures must be supplied by the authorised persons.



Download FoodData Central Data

Documentation

Data contained in FoodData Central can be downloaded. The download files are available both as an Excel compatible CSV delimited ASCII file and in JavaScript Object Notation (JSON).

View data type documentation for each of FoodData Central's data types. Documentation for the JSON format can be found through our API guide on Swagger Hub.

View Download Field Descriptions (PDF – 1MB) for definitions of all the terms and labels used in the download files of FoodData Central. This guide is also included in each downloaded file.

Latest Downloads

Releases

Data Type	Release Date	Download File	File Format	Zipped	Unzipped
Foundation Foods	04/2026	April 2026 (JSON) April 2026 (CSV)	JSON CSV	993K 3.7M	6.5M 30M
SR Legacy	04/2018	April 2018 (JSON) April 2018 (CSV)	JSON CSV	12.2M 5.7M	202M 54M
FNDDS	10/2024	FNDDS 2023-2023 (JSON) FNDDS 2023-2023 (CSV)	JSON CSV	3.7M 200K	64M 1.4G
Branded	04/2026	April 2026 (JSON) April 2026 (CSV)	JSON CSV	190M 426M	3.1G 2.1G
Full Download of All Data Types	04/2026	April 2026 (CSV)	CSV	469M	3.1G

Historical Downloads

Supporting Data

Starting in April 2023, all downloads now include the supporting data in every archive. These supporting data is only required for archives prior to the April 2023 releases.

Data Type	Release Date	Download File	File Format	Zipped	Unzipped
Supporting Data	10/2022	October 2022 (CSV)	CSV	352K	16M
Supporting Data	4/2022	April 2022 (CSV)	CSV	1.1M	17.6M
Supporting Data	10/2021	October 2021 (CSV)	CSV	1.1M	17.6M
Supporting Data	4/2021	April 2021 (CSV)	CSV	1.1M	17.6M
Supporting Data	10/2020	October 2020 (CSV)	CSV	1.1M	17.6
Supporting Data	4/2020	April 2020 (CSV)	CSV	1.6M	25M
Supporting Data	12/2019	December 2019 (CSV)	CSV	210K	291K
Supporting Data	4/2019	April 2019 (CSV)	CSV	212K	340K

Releases

Data Type	Release Date	Download File	File Format	Zipped	Unzipped
Foundation Foods	12/2025	December 2025 (JSON) December 2025 (CSV)	JSON CSV	467K 2.4M	6.0M 25M
Foundation Foods	04/2025	April 2025 (JSON) April 2025 (CSV)	JSON CSV	454K 2.3M	5.9M 24M
Foundation Foods	10/2024	October 2024 (JSON) October 2024 (CSV)	JSON CSV	446K 2.3M	6M 24M
Foundation Foods	04/2024	April 2024 (JSON) April 2024 (CSV)	JSON CSV	446K 2.3M	5.7M 24M
Foundation Foods	10/2023	October 2023 (JSON) October 2023 (CSV)	JSON CSV	409K 2.3M	5.5M 24M
Foundation Foods	4/2023	April 2023 (JSON) April 2023 (CSV)	JSON CSV	391K 2.3M	5.2M 23M
Foundation Foods	10/2022	October 2022 (JSON) October 2022 (CSV)	JSON CSV	366K 2.4M	5.1M 22.6M
Foundation Foods	4/2022	April 2022 (JSON) April 2022 (CSV)	JSON CSV	356K 2.3M	5.1M 23M
Foundation Foods	10/2021	October 2021 (JSON) October 2021 (CSV)	JSON CSV	317K 2.1M	4.2M 19M
Foundation Foods	4/2021	April 2021 (CSV)	CSV	1.8M	16M
Foundation Foods	10/2020	October 2020 (CSV)	CSV	1.8M	14M
Foundation Foods	4/2020	April 2020 (CSV)	CSV	2.3M	14.2M
Foundation Foods	12/2019	December 2019 (CSV)	CSV	1.6M	13.2M
Foundation Foods	4/2019	April 2019 (CSV)	CSV	1.5M	10.5M
FNDDS	10/2021	FNDDS 2021-2021 (JSON) FNDDS 2021-2021 (CSV)	JSON CSV	5.7M 200K	64M 1.4G
FNDDS	10/2022	FNDDS 2021-2021 (JSON) FNDDS 2021-2021 (CSV)	JSON CSV	3.8M 4.3M	46.6M 41M
FNDDS	10/2021	FNDDS 2017-2018 (JSON) FNDDS 2017-2018 (CSV)	JSON CSV	4.2M 5M	64.3M 54M
FNDDS	4/2020	FNDDS 2015-2016 (CSV)	CSV	5M	41M
FNDDS	4/2018	FNDDS 2013-2014 (CSV)	CSV	5M	38M
Branded	12/2025	December 2025 (JSON) December 2025 (CSV)	JSON CSV	195M 427M	3.1G 2.0G
Branded	04/2025	April 2025 (JSON) April 2025 (CSV)	JSON CSV	195M 423M	3.1G 2.0G
Branded	10/2024	October 2024 (JSON) October 2024 (CSV)	JSON CSV	195M 420M	3.1G 2.0G
Branded	04/2024	April 2024 (JSON) April 2024 (CSV)	JSON CSV	178M 415M	3.1G 2.0G
Branded	10/2023	October 2023 (JSON) October 2023 (CSV)	JSON CSV	174M 408M	3.1G 1.9G
Branded	4/2023	April 2023 (JSON) April 2023 (CSV)	JSON CSV	170M 397M	2.8G 1.9G
Branded	10/2022	October 2022 (JSON) October 2022 (CSV)	JSON CSV	164M 383M	2.7G 1.8G
Branded	4/2022	April 2022 (JSON) April 2022 (CSV)	JSON CSV	163M 379M	2.6G 1.8G
Branded	10/2021	October 2021 (JSON) October 2021 (CSV)	JSON CSV	157M 362M	2.6G 1.8G
Branded	4/2021	April 2021 (CSV)	CSV	221M	1.6G
Branded	10/2020	October 2020 (CSV)	CSV	96M	663M
Branded	4/2020	April 2020 version 2* (CSV)	CSV	78M	487M
Branded	12/2019	December 2019 (CSV)	CSV	68M	430M
Branded	4/2019	April 2019 (CSV)	CSV	58M	357.5M
Branded (before the Launch of FoodData Central)	7/2018	July 2018 (CSV) July 2018 (MS Access)	CSV MS Access	38M 79M	272M 536M
Full Download of All Data Types	12/2025	December 2025 (CSV)	CSV	458M	3.1G
Full Download of All Data Types	04/2025	April 2025 (CSV)	CSV	453M	3.1G
Full Download of All Data Types	10/2024	October 2024 (CSV)	CSV	446M	3.1G
Full Download of All Data Types	04/2024	April 2024 (CSV)	CSV	446M	2.8G
Full Download of All Data Types	10/2023	October 2023 (CSV)	CSV	413M	2.8G
Full Download of All Data Types	4/2023	April 2023 (CSV)	CSV	397M	2.8G
Full Download of All Data Types	10/2022	October 2022 (CSV)	CSV	377M	2.7G
Full Download of All Data Types	4/2022	April 2022 (CSV)	CSV	372M	2.7G
Full Download of All Data Types	10/2021	October 2021 (CSV) October 2021 (MS Access)	CSV MS Access	316M 101M	2.4G 1.4G
Full Download of All Data Types	4/2021	April 2021 (CSV) April 2021 (MS Access)	CSV MS Access	224M 168M	1.7G 920M
Full Download of All Data Types	10/2020	October 2020 (CSV) October 2020 (MS Access)	CSV MS Access	192M 268M	760M 1.7G
Full Download of All Data Types	4/2020	April 2020 version 2* (CSV) April 2020 version 2* (MS Access)	CSV MS Access	93M 153M	626M 962M
Full Download of All Data Types	12/2019	December 2019 (CSV) December 2019 (MS Access)	CSV MS Access	68M 120M	465M 765M

Figure A.2: USDA FoodData Central download portal (source screenshot; not an application UI).

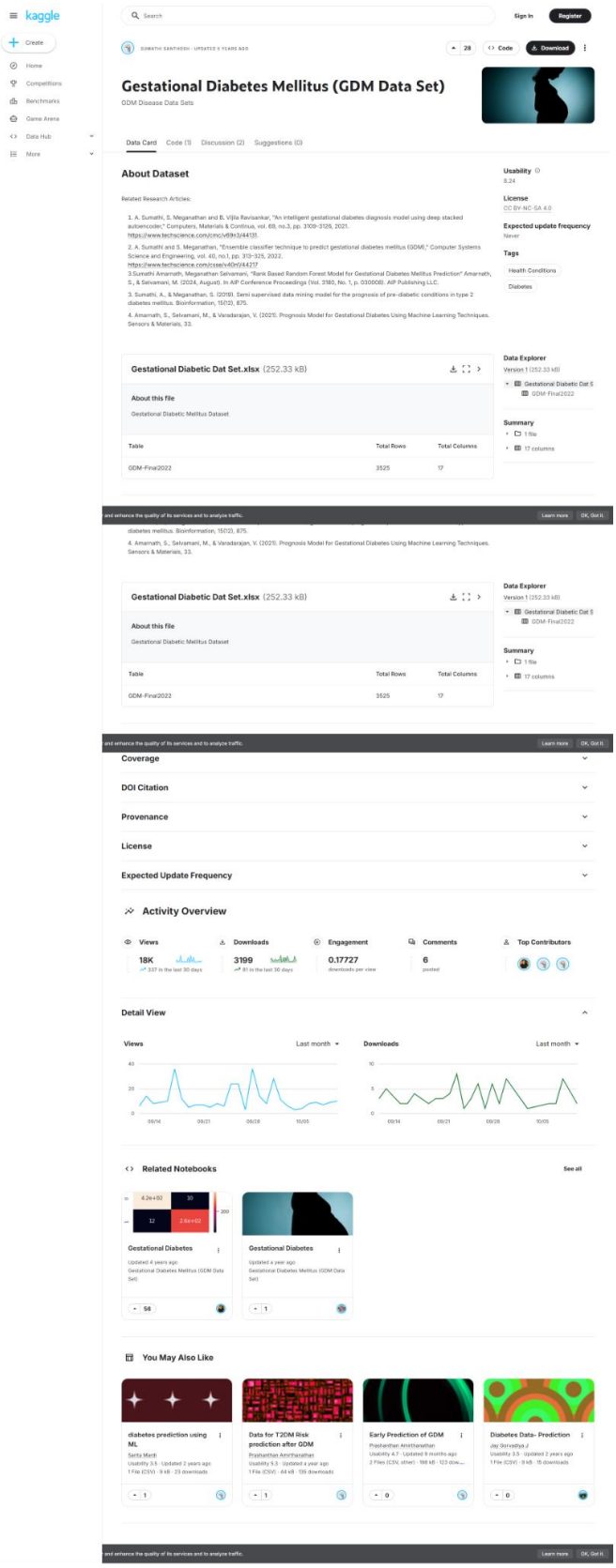


Figure A.3: Kaggle GDM dataset landing page (source screenshot; not the YOLO food-image dataset).

Appendix A.1 — Reported result versus independent verification

Metric or evidence	Source status	What remains needed
YOLO mAP@50 = 0.9392	Student-reported metric	Original training/evaluation export
Dataset images, classes and splits	Description only	Dataset manifest and split audit
Supabase schema	Developer code-inspection report	SQL migrations / redacted schema export
FAO/USDA composition data	Official sources verified	Actual food-to-reference mapping
GDM warning thresholds	Reported app logic	Clinician review; negative tests
App interface screenshots	Not supplied	Capture from actual running app

Appendix A.2: Supplied model and data artefacts

Two checkpoint files, best (3).pt and best (4).pt, were provided. They have different file sizes; the supplied evidence does not establish which one generated the reported validation screenshots. A credible replication should record the SHA-256 fingerprint of the chosen file, the precise training dataset YAML, Ultralytics and Python versions, image size, split, and validation command. These files are research artefacts, not evidence of approved clinical-device deployment.

The project materials additionally include a maternal_food_db spreadsheet, maternal/GDM criteria spreadsheets, a Maathru Care presentation, and a code-inspection transcript. Their existence is documented; the underlying clinical thresholds, input-unit definitions and food composition need independent quality review. A separate WiDS Datathon 2021 diabetes notebook concerns ICU patients and general diabetes, not the same pregnancy-specific cohort, and must not be cited as a GDM training experiment conducted for this nutrition component.

Appendix A.3: Figure provenance and interpretation

Figures 6.1–6.3 are original researcher-provided screenshots/outputs, not regenerated results. Figure 6.1 gives aggregate object-detection metrics. Figure 6.2 displays class-normalised detection confusions. Figure 6.3 is a selection of prediction examples. None independently verifies patient outcomes, dietary counselling safety, or food nutrient ground truth.

Appendix B Academic integrity and AI use disclosure

Academic integrity and AI-use disclosure: Generative AI was used to assist in language editing, restructuring an earlier thesis draft, interpreting a user-supplied code-inspection report, drafting tables and formatting. It was not used here to fabricate participant data, YOLO training outputs, nutritional benchmark results or clinical validation. The reported mAP@50 of 0.9392 is a student-provided figure pending source-log verification. The student remains responsible for checking all technical assertions, reference details and the institute's current AI-use requirements. This statement should be checked against the actual tools used during the research and the university policy before it is signed.

Research and reporting ownership remains with Nuwantha P.A.T. (IT22284716). Any additional AI services used to generate code, labels or model training scripts, and any human reviewer feedback, must be disclosed accurately as appropriate to the SLIIT course instructions. Signatures and the co-supervisor field must be completed by the authorised people; none have been invented.

Appendix C Source provenance and version notes

Source provenance: The original March proposal named a broad Virtual Nutritionist and GDM Action Engine with Flutter/Firebase among the proposed technologies. The subsequent student-confirmed implementation initially focused on ingredient-based Sri Lankan meal suggestions and nutritional analysis. A later developer-supplied source-code inspection describes a React Native/Expo client, a custom YOLO food detection microservice, Supabase/PostgreSQL nutrition tables and GDM warning logic. This revision documents those additions without treating the source-code transcript as a reproducible original experiment. Validation screenshots and model checkpoint files are now included; complete training logs, current backend code and database export remain outside the reviewed file set.